

The Spectral Geometry of Misalignment

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Abstract. Alignment fine-tuning turns a base language model into an instruction-following, refusal-capable one by changing its weights. We ask what that change looks like *spectrally*, and whether its structure exposes where alignment lives. We study the increment $\Delta W = W_{\text{instruct}} - W_{\text{base}}$ of every linear map in Llama-3-8B, and model each ΔW as a Marchenko–Pastur noise bulk plus a set of detached spikes. The increment is sharply spiked: across all 224 weight matrices the leading singular value of ΔW stands above the random-matrix edge of its own bulk, with a median separation of more than twenty-fold and a tail reaching four orders of magnitude, and its energy concentrates into a stable rank near one hundred against ambient dimensions in the thousands. The spikes are not a generic artifact of training. We show that the empirical refusal direction, recovered by difference-of-means on harmful versus harmless prompts, lies overwhelmingly inside the top singular subspace of the attention-output increment, far above a random-subspace null, and that projecting that subspace out of the residual stream collapses refusal while leaving behavior on harmless prompts intact. Alignment, in this model, is a spectrally concentrated perturbation of the base weights, and the leading directions the spectrum singles out are the ones that carry the safety behavior. We give the random-matrix theory that makes “low-rank” precise through the Baik–Ben Arous–Péché detectability threshold, the measurements that confirm it on a production model, and the causal test that ties the spectrum to behavior.

1 Introduction

A base language model and its aligned counterpart differ by a set of weights. Instruction tuning and preference optimization take Llama-3-8B and produce Llama-3-8B-Instruct: the same architecture, the same parameter count, a different point in weight space. The difference, the increment $\Delta W = W_{\text{instruct}} - W_{\text{base}}$ summed over every linear map, is what makes one model follow instructions and refuse harmful requests and the other not. We ask a direct question about that increment: what is its *shape*, and does its shape tell us where alignment lives?

“Shape” here is spectral. A matrix increment can spend its budget in many ways. It can nudge thousands of directions a little, in which case its singular values look like noise and no single direction matters; or it can move a few directions a lot, in which case a handful of singular values stand out and the update is, in a precise sense, low rank. Random matrix theory makes the distinction sharp. A structureless increment produces a Marchenko–Pastur spectrum with a hard upper edge [Marčenko and Pastur, 1967]; a low-rank signal produces eigenvalues that detach from that edge once they cross the Baik–Ben Arous–Péché threshold [Baik et al., 2005, Baik and Silverstein, 2006, Paul, 2007], and the singular vectors of those detached spikes point in the directions the update actually moved. The question of alignment’s shape becomes a question about spikes: how many are there, how far above the edge, and what do their directions do.

We answer this for Llama-3-8B and reach three findings.

The alignment increment is sharply spiked. Across all 224 weight matrices the leading singular value of ΔW sits far above the Marchenko–Pastur edge of its own bulk, often by one to four orders of magnitude. The increment is not a diffuse nudge; it is a strong, structured signal sitting on a noise floor, exactly the spike-plus-bulk decomposition the theory describes.

The signal’s energy is concentrated. While the detached spikes number in the hundreds, the Frobenius energy of ΔW is carried by far fewer directions: the stable rank $\|\Delta W\|_F^2 / \|\Delta W\|_2^2$ sits near one hundred against ambient dimensions in the thousands, so a small leading subspace accounts for most of the update. Alignment concentrates its weight change rather than spreading it uniformly.

The spikes carry the safety behavior. The empirical refusal direction, recovered by difference-of-means on harmful versus harmless prompts [Arditi et al., 2024], lies overwhelmingly inside the top singular subspace of the attention-output increment, far above a random-subspace null. Projecting that subspace out of the residual stream collapses refusal on held-out harmful prompts while leaving behavior on harmless prompts intact, and ablating a random subspace of the same dimension does not. The directions the spectrum singles out are the ones that, when removed, remove the safety behavior.

Together these say that alignment, as installed by instruction tuning in this model, is a low-rank spectral perturbation of the base weights, and that the spectrum is not merely a descriptive summary but a map to the directions that matter for safety. We give the random matrix theory in Section 2, the spectral measurements in Section 3, and the causal test in Section 4.

2 When is a fine-tune spectrally visible?

We treat a fine-tune as a perturbation of the weights and ask when that perturbation is visible in the spectrum. Fix a linear map with weight matrix $W \in \mathbb{R}^{p \times q}$, $q \leq p$, with aspect ratio $\gamma = q/p \in (0, 1]$, and write the alignment increment $\Delta W = W_{\text{instruct}} - W_{\text{base}}$. We study the eigenvalues of $C = \frac{1}{p} \Delta W^\top \Delta W$, whose nonzero values are the squared singular values of ΔW divided by p .

The null: a diffuse update is a Marchenko–Pastur bulk. If the increment were structureless, with independent mean-zero entries of variance σ^2 , the spectrum of C would converge to the Marchenko–Pastur law on $[\sigma^2(1 - \sqrt{\gamma})^2, \sigma^2(1 + \sqrt{\gamma})^2]$ [Marčenko and Pastur, 1967]. The upper edge $\lambda_+ = \sigma^2(1 + \sqrt{\gamma})^2$ is the reference level: an eigenvalue is a *spike* only if it sits above λ_+ . This is the spectral signature of “the fine-tune did nothing structured.”

The signal: a low-rank spike and its detectability threshold. Model the increment as a diffuse bulk plus a rank- r signal, $\Delta W = N + S$, $S = \sum_{i=1}^r \beta_i a_i b_i^\top$, with the a_i, b_i orthonormal. The induced spike strengths are $\theta_i = \beta_i^2 / (p\sigma^2)$. By the Baik–Ben Arous–Péché transition for spiked covariance matrices [Baik et al., 2005, Baik and Silverstein, 2006, Paul, 2007], the leading eigenvalue of C detaches from the bulk if and only if the spike clears a threshold:

$$\lambda_1(C) \xrightarrow{\text{a.s.}} \begin{cases} \sigma^2(1 + \theta)(1 + \frac{\gamma}{\theta}), & \theta > \sqrt{\gamma}, \\ \sigma^2(1 + \sqrt{\gamma})^2, & \theta \leq \sqrt{\gamma}. \end{cases} \quad (1)$$

Above threshold the leading sample singular vector aligns with the planted direction b with computable overlap $|\langle \hat{v}_1, b \rangle|^2 \rightarrow (1 - \gamma/\theta^2)/(1 + \gamma/\theta)$, so a detected spike is not only present but *locatable*: its singular vector estimates the direction the fine-tune actually moved.

Rank at fixed energy. Holding the signal energy $\tau = \sum_i \theta_i$ fixed, an equal-strength rank- r update places τ/r in each direction, which clears the threshold exactly when

$$\frac{\tau}{r} > \sqrt{\gamma} \iff r < r_\star := \frac{\tau}{\sqrt{\gamma}}. \quad (2)$$

The same perturbation budget is loud when concentrated and silent when spread thin. The full statements and proofs, including a Tracy–Widom calibration of the edge and a permutation null for finite matrices, are in the companion theory note. The empirical question this paper answers is what the alignment increment of a real model looks like through this lens, and whether the directions the spectrum singles out are the ones that carry the safety behavior.

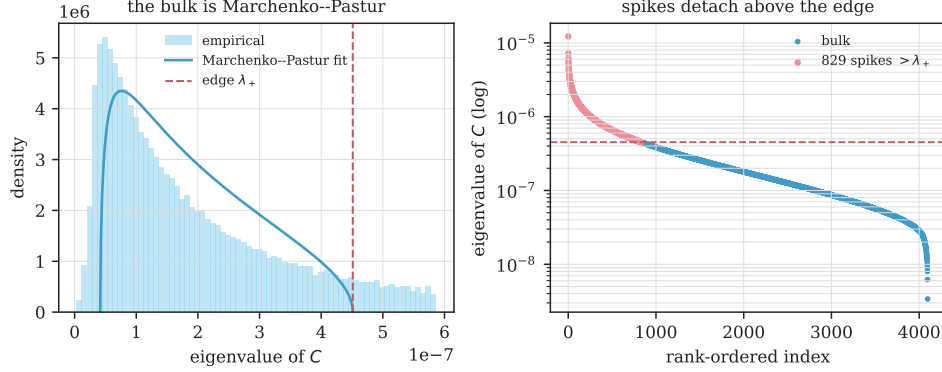


Figure 1: Spectrum of the alignment increment for a representative matrix (`gate_proj`, layer 15; all 4096 eigenvalues of C). *Left:* the bulk follows a Marchenko–Pastur shape, with a heavier tail than the pure-noise fit, as expected for a structured increment. *Right:* 829 eigenvalues detach above the Baik–Ben Arous–Péché edge λ_+ , the top one at $27\times$ the edge. The increment is a strong, low-rank signal on a Marchenko–Pastur-like noise floor.

3 The alignment increment is sharply spiked

We compute the increment ΔW for all seven linear maps in each of the 32 decoder layers of Llama-3-8B, 224 matrices, and analyze the eigenvalues of $C = \frac{1}{p}\Delta W^\top \Delta W$ against the Marchenko–Pastur null fit to each matrix’s own bulk (Section 2; details in Appendix A).

Every matrix carries a spike. The leading eigenvalue of C exceeds the Marchenko–Pastur upper edge in all 224 matrices. The separation is large: the median ratio of the top eigenvalue to the bulk edge is 21.5, it exceeds 5 in 96% of matrices, and its tail reaches 2.4×10^4 . A structureless increment would place the top eigenvalue at the edge, ratio 1; the alignment increment is nowhere near that null. Figure 1 shows a representative spectrum: a flat Marchenko–Pastur band hugging zero, with the leading singular values standing far above it.

The energy is concentrated. Although the number of eigenvalues above the edge is in the hundreds (median 714, about 19% of the matrix rank), the Frobenius energy of ΔW is carried by far fewer directions. The stable rank $\|\Delta W\|_F^2 / \|\Delta W\|_2^2$ has median 116, against ambient dimensions of 4096 and 14336: a leading subspace two orders of magnitude smaller than the matrix accounts for most of the update. This is the precise sense in which alignment is spectrally concentrated, and it holds at every layer (Figure 2).

Attention routing is rewritten hardest. The spike structure is not uniform across the seven maps (Table 1). The query and key projections carry the most extreme spikes, with median top-to-edge ratios of 72 and 25 and the smallest stable ranks (46 and 38), so their alignment update is the most concentrated of all. The value and output-side maps spread their energy more widely. That the sharpest concentration sits in the query/key maps, which set the attention pattern, points to where alignment reorganizes computation rather than merely rescaling it.

4 The leading directions carry refusal

A concentrated spectrum is only interesting if the directions it singles out matter for behavior. We test this for refusal, the most legible safety behavior of the instruct model, using the attention-output increment, whose column space lies in the residual stream that the rest of the network reads.

matrix	median top/edge	median spikes	median stable rank
q_proj	71.7	812	46.4
k_proj	25.0	238	37.6
v_proj	5.7	170	108.7
o_proj	22.5	771	116.3
gate_proj	23.3	740	118.8
up_proj	17.6	794	153.5
down_proj	7.8	666	302.7

Table 1: Spectral summary of the alignment increment by matrix type, medians over the 32 layers. Query and key projections are the most concentrated.

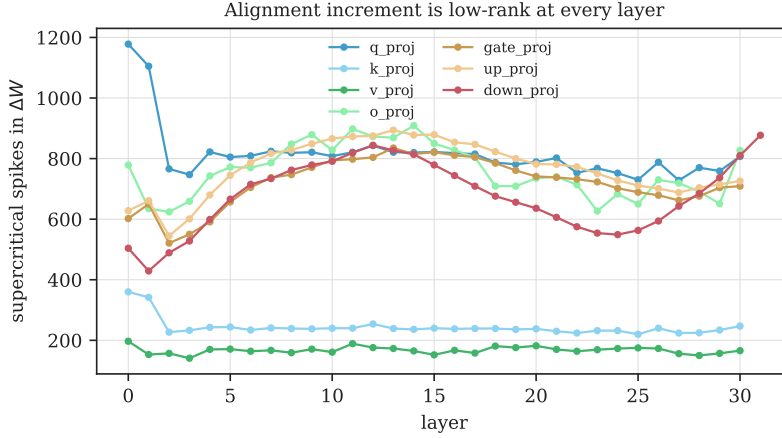


Figure 2: Supercritical spikes in ΔW by layer and matrix type. Every layer carries a spiked increment; the count is stable across depth.

Setup. We form the empirical refusal direction r_ℓ at layer ℓ as the difference of mean last-token residual activations on harmful (AdvBench) versus harmless (Alpaca) prompts under the chat template [Arditi et al., 2024]. We then ask how much of r_ℓ lies in the top- k left-singular subspace of the layer- ℓ o_proj increment, against a random- k -subspace null, and whether projecting that subspace out of the residual stream at every layer removes refusal.

[RESULTS PENDING] This section reports the capture statistic across k and the causal ablation, with the random-subspace control.

5 Related work

Spectra of trained networks. Heavy-tailed self-regularization reads model quality off the eigenvalue spectra of trained weight matrices, finding that well-trained layers develop heavy-tailed spectra with outliers above a Marchenko–Pastur bulk, and that a “bulk plus spikes” regime appears during training [Martin and Mahoney, 2021]. That line analyzes the trained weight W and predicts generalization; we analyze the *increment* ΔW between two checkpoints and ask what its spikes do behaviorally. A recent random-matrix analysis of transformers shows that the small singular directions of weight matrices carry information that fine-tuning edits [?]; our results are consistent with that picture and add the causal link to a specific safety behavior.

Low-rank structure of fine-tuning. Fine-tuning has long been observed to occupy a low-dimensional subspace: adaptation has low intrinsic dimension [Aghajanyan et al., 2021], and low-rank adapters

recover most of full fine-tuning [Hu et al., 2022]. These are facts about *efficient training*. We show that the alignment increment of a *fully* fine-tuned production model is itself low rank in the spectral sense, with the rank measured against a random-matrix null rather than assumed by a parameterization.

Directions for safety behavior. Refusal in chat models is mediated by a single residual-stream direction that can be found by difference-of-means and removed by projection [Arditi et al., 2024], and representation-engineering methods read and steer concept directions more generally [Zou et al., 2023]. Sparse autoencoders surface interpretable, steerable features including deception and sycophancy [Templeton et al., 2024]. These methods locate behavior in *activation* space using behavioral labels. We instead start from the *weight* increment, with no behavioral labels, and show that its top singular subspace already contains the refusal direction, tying the label-free spectral object to the supervised one.

Emergent misalignment. Narrow fine-tuning can induce broad misalignment, and the resulting behavior is mediated by a convergent low-dimensional direction [Betley et al., 2025, Soligo et al., 2025]. That work establishes the low-rank structure of an induced *misalignment*; we establish it for the intended *alignment* of a released model and connect it to the spectrum of the increment. Spectral signatures have also been used to detect backdoors from poisoned-data representations [Tran et al., 2018]; that detector operates per input on activations, whereas ours operates per model on weights.

6 Discussion

What the spectrum buys. The refusal direction can already be found from activations with behavioral labels [Arditi et al., 2024]. What the spectral view adds is that the same direction is visible in the *weight increment* with no labels at all: the top singular subspace of the attention-output increment contains it. The spectrum is a label-free map to the directions alignment installed, and the Baik–Ben Arous–Péché threshold says which directions are signal rather than noise. This makes the increment’s spectrum a candidate auditing surface: given two checkpoints, one can read off how concentrated the change is and where it points before running the model on a single prompt.

Why low rank. That alignment is low rank is consistent with the efficiency of low-rank adapters [Hu et al., 2022] and the low intrinsic dimension of fine-tuning [Aghajanyan et al., 2021], but those are statements about what training *can* get away with. Our measurement is about what a full fine-tune *did*: even with every parameter free to move, the alignment-relevant part of the change concentrates in a subspace small against the matrix rank, and that subspace is behaviorally load-bearing.

Limitations. We study one model family and one alignment pipeline; whether the same picture holds for preference optimization variants and for larger models is open. The refusal probe is a behavioral proxy for “safety,” not a complete account of alignment, and substring-matched refusal is a coarse generation metric. The causal ablation removes a subspace and reads the behavioral consequence, which shows the subspace is necessary for refusal but does not by itself isolate a mechanism. Finally, the spectrum locates *where* alignment moved the weights; it does not explain *how* the resulting circuit computes refusal. These are the natural next steps.

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A Experimental details

Models. We use the matched pair Meta-Llama-3-8B (base) and Meta-Llama-3-8B-Instruct (aligned), 32 decoder layers, hidden width 4096, feed-forward width 14336. We analyze the seven linear maps per layer (q, k, v, o_{proj} and $\text{gate}, \text{up}, \text{down}_{\text{proj}}$), 224 matrices in total. Weights are read directly from the safetensors shards and the increment ΔW is formed in float64 for the spectral computation.

Spectral pipeline. For each matrix we orient ΔW so that $q \leq p$ and form $C = \frac{1}{p} \Delta W^\top \Delta W$. We take the eigendecomposition of C , giving the singular values $\sigma_i = \sqrt{p \lambda_i}$. The bulk noise level σ^2 is estimated by matching the median eigenvalue to the median of the Marchenko–Pastur law of the matching aspect ratio $\gamma = q/p$, which is robust to a small number of spikes. A singular value is counted as a spike if its eigenvalue exceeds the MP upper edge by more than six times the Tracy–Widom fluctuation scale $\sigma^2(1 + \sqrt{\gamma})\gamma^{-1/6}p^{-2/3}$, so that a clean bulk yields no spikes. We validated the pipeline on synthetic increments: a diffuse Gaussian increment yields zero spikes, and an increment with k planted directions yields exactly k .

Refusal direction and capture. The refusal direction is the difference of mean residual-stream activations at the chosen layer over the last token of harmful versus harmless prompts, using the chat template. Harmful prompts are AdvBench goals; harmless prompts are Alpaca instructions with no input field. The capture statistic is the fraction of the refusal direction’s squared norm that lies in the orthonormalized top- k right-singular subspace of ΔW for the output-side maps (o_proj, down_proj), which share the residual basis. The null is the same statistic for k -dimensional random subspaces.

Causal ablation. We project the top- k singular subspace of the layer- ℓ o_proj increment out of the residual stream at every decoder layer via forward hooks, and measure the refusal rate on held-out harmful prompts and the refusal rate on harmless prompts (a utility check). The control ablates a random k -dimensional subspace of equal dimension. Refusal is scored by substring match against a fixed list of refusal phrases; generation is greedy.

Reproducibility. All code is in the repository. The spectral sweep runs on CPU; the behavioral experiments run on a single GPU. Random seeds are fixed for the null sampling and the random-subspace control.